INTEREST RATE RISK IN THE BANKING BOOK

Modeling non-maturity deposits

What Basel, the CRD/CRR and the EBA require, how banks build the models, and how the hedges are accounted for

90-minute core path + deep-dives

ROADMAP

Fourteen parts: a 90-minute core path

#	Topic	Core	Extra
1	Why NMDs matter	9′	
2	The regulatory stack	2′	1
3	Basel principles and SA	4′	
4	EBA GL and the 5-year cap	8′	
5	Data and segmentation	2′	2
6	Volume models	9′	4
7	Rate models	9′	2

#	Topic	Core	Extra
8	From model to EVE and NII	9′	3
9	Validation and governance	2′	3
10	EU standardized approach	5′	3
11	Supervisory outlier tests	4′	1
12	Evidence and 2023 case	4′	3
13	Hedging and IAS 39 carve-out	13′	3
14	Debates, takeaways, exercises	6′	3

> Core: minutes in the 90-minute lecture. Extra: deep-dive slides for a second session or self-study.

LEARNING OBJECTIVES

By the end you should be able to

01

Explain the problem

Why a deposit repayable on demand can behave like long-term, low-rate funding

02

Map the rulebook

Which Basel, CRD, CRR, RTS and EBA texts apply, and what each one does

03

Build the models

Volume, rate and migration models, and the repricing profile they produce

04

Apply the SA

Split a deposit book into core and non-core under the EU caps

05

Account for the hedge

Explain the IAS 39 EU carve-out and what may replace it

06

Challenge a model

Judge assumptions the way a supervisor or validator would

PART 1 · WHY NMDS MATTER

Non-maturity deposits fund most EU banks

>50%

of liabilities (ex derivatives) are NMDs for half of the banks in the EBA sample

65%

average share of all deposits in banking-book liabilities

76%

of those deposits have no maturity date

Mix of NMDs (mean share of total NMD)



Source: EBA/REP/2025/04, QIS and ITS data as of December 2023 · 4

PART 1 · THE MODELING PROBLEM

Overnight by contract, long by behavior

Contractually

- Withdrawable at any time
- Rate set by the bank, changeable at will
- Repricing date: overnight

Behaviorally

- Balances are sticky in aggregate
- Client rates follow market rates slowly and partially
- Effective repricing: often years

Volume

How much of the balance stays, and for how long?

Rate

How much of a market move is passed to clients, and how fast?

Migration

Do clients move into term deposits or other products when rates rise?

PART 1 · WHY THE ASSUMPTION MATTERS

Long deposit models justify long assets

Model NMDs as multi-year, low-rate funding → Hold fixed-rate loans and bonds as a "natural hedge" → If deposits leave or reprice faster: EVE and NII losses

> March 2023: fast deposit outflows at several banks showed that balances can be far less sticky than models assumed.
>
> The BCBS and EBA follow-up work on IRRBB since then focuses heavily on NMD assumptions.

PART 2 · THE REGULATORY STACK

Three layers of rules for IRRBB

Basel Committee

Global standard, not law

- IRRBB standard (2016), now SRP31
- Application guidance SRP98
- Shock recalibration (July 2024)

EU Level 1

Directive and Regulation

- CRD Art. 84: internal systems, standardized approach
- CRD Art. 98(5), (5a): outlier tests, supervisory powers
- CRR Art. 448: Pillar 3 disclosure

EU Level 2 and 3

RTS, ITS and guidelines

- RTS 2024/856: outlier tests
- RTS 2024/857: standardized methods
- ITS 2024/855: reporting
- EBA GL 2022/14: IRRBB and CSRBB

Supervisory monitoring: EBA IRRBB heatmap (Jan 2024), implementation reports (Feb 2025, Jan 2026), Single Rulebook Q&As

PART 2 · HOW WE GOT HERE

A decade of IRRBB rule-making

Apr 2016
BCBS IRRBB standard (d368)

2019
CRD V: NII outlier test, mandates for EBA RTS and GL

Oct 2022
EBA package: GL 2022/14 and draft RTS

Jun 2023
GL 2022/14 apply, including the 5-year NMD cap

Nov 2023
EBA Q&A 2023_6807 clarifies the cap

Jan 2024
EBA IRRBB heatmap sets a monitoring plan

May 2024
RTS 2024/856 and 2024/857 in force (14 May)

Jul 2024
BCBS recalibrates the shocks (d578)

Feb 2025
EBA report: NMD assumptions, margins, hedging

Jan 2026
Second EBA report: keep the 5-year cap

Top row: standards and guidelines. Bottom row: technical standards and monitoring · 8

PART 3 · BASEL PRINCIPLES

Four BCBS principles shape NMD modeling

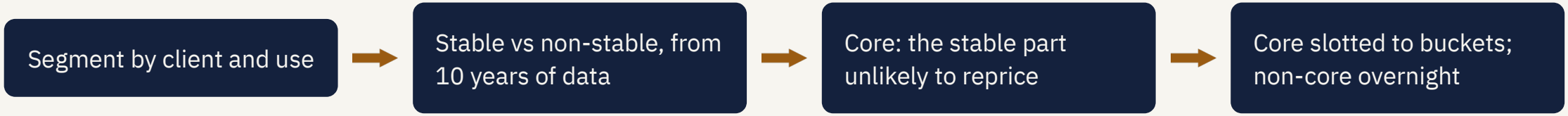
Principle	What it requires for NMDs
4 Measurement	Measure IRRBB on both economic value (EVE) and earnings (NII) bases
5 Assumptions	Behavioral assumptions, NMDs first among them, must be understood, conceptually sound, documented, rigorously tested and aligned with strategy
8 Disclosure	Publish IRRBB levels, including the average and longest repricing maturity assigned to NMDs
12 Outlier test	Supervisors flag banks whose EVE falls by more than 15% of Tier 1 under the prescribed shocks

> Banks may use their own models. The standardized framework is a fallback that supervisors can require.

Source: BCBS, Interest rate risk in the banking book (d368, 2016), now SRP31 · 9

PART 3 · BASEL STANDARDIZED FRAMEWORK

Segment, split, cap, slot



Category	Max core share	Max average maturity of core
Retail transactional	90%	5 years
Retail non-transactional	70%	4.5 years
Wholesale	50%	4 years

Source: BCBS SRP31; the EU copied these caps into RTS 2024/857 Art. 8 · 10

PART 4 · EBA GL 2022/14

What an internal NMD model must show

Proportionate

Set in line with size, complexity and business strategy

para 103

Rate elasticity

How client rates adjust to market rates

para 108(b)

5-year cap

Maximum weighted average repricing date for NMDs

para 111

Scenario-dependent

Behavior adjusted to each rate scenario

also RTS 2024/856 Art. 3(2)

Tested

Sensitivity of results to key assumptions, back-testing

model governance

Documented

Rationale, data and expert judgment written down

Basel Principle 5

PART 4 · THE ONE HARD NUMBER IN THE GUIDELINES

5 years

Maximum weighted average repricing date for NMDs in the bank's internal measurement system

Scope

Retail, wholesale non-financial, and operational deposits of financial clients

Measured on

The whole portfolio, core and non-core together, separately per currency

Deviation

Only if the bank shows the supervisor unintended, counterintuitive effects

Source: EBA/GL/2022/14 para 111; EBA Q&A 2023_6807 (Nov 2023) · 12

PART 4 · ANATOMY OF AN NMD MODEL

Three sub-models feed one repricing profile

Volume model

Stable vs volatile balances; run-off or decay curve

Rate model

Pass-through (beta), lag, asymmetry up vs down, zero floor

Migration model

Switching into term deposits and other products



Output

Core / non-core split per scenario

Core slotted into repricing buckets

Projected client rate and margin for NII

PART 4 · INDUSTRY PRACTICE

The rules prescribe properties, not a method

Approach	What it estimates	Watch-outs
Replicating portfolio	Mix of tenors whose rolling yield best tracks the deposit margin	Optimizes margin stability, not risk; fitted on past rates
Decay or survival curves	How fast balances run off, by vintage or segment	Needs long data spanning rate cycles; SA asks for 10 years
Pass-through regression	Long-run beta and speed of adjustment, e.g. error-correction models	Low-rate history understates beta; asymmetry up vs down
Expert overlays and stress	Adjustments for digital banking, concentration, new products	Must be documented and challenged; reverse-stress the overlay

> Whatever the method, the output must respect the 5-year cap and change with the scenario.

PART 5 · DATA AND SEGMENTATION

What you need before you model anything

The data set

- Monthly, ideally daily, balances by account or segment
- Client rates paid, by product and date
- Market rates: €STR, Euribor, swap curve
- Account opening and closing dates
- Client and product attributes: type, channel, deposit-guarantee coverage
- Flows into term deposits and other products

What to clean

- Structural breaks: mergers, product redesigns, book transfers
- Exceptional periods: negative rates 2014–22, pandemic savings 2020–21
- Seasonality: salaries, tax dates, corporate year-end
- Very large single depositors: model separately
- Survivorship: closed accounts must stay in the data

> At least one full rate cycle; the standardized approach asks for 10 years of volume history.

10-year requirement: RTS 2024/857 Art. 8(2)(a) · 15

PART 5 · SEGMENTATION

Segment so each group behaves alike

Dimension	Why it matters	Typical split
Regulatory category	Needed for SA and reporting	Retail transactional / non-transactional, wholesale non-financial / financial
Remuneration	Rate sensitivity	Zero-rate vs interest-bearing
Deposit insurance	Run risk	Within the €100k guarantee vs above it
Balance size	Wealth, rate awareness	Bands, e.g. under €10k, €10–100k, over €100k
Channel	Speed of switching	Branch clients vs digital-only clients
Corporate use	Cash-management needs	Operational balances vs excess liquidity

> Good segments differ in decay and beta, stay stable over time, and keep enough data in each cell.

EU deposit guarantee: €100,000 per depositor per bank (Directive 2014/49/EU) · 16

PART 5 · LEVEL OF MODELING

Portfolio models or account-level models

Portfolio (aggregate)

Data: total balance per segment over time

Methods: volatility envelope, balance regressions, error-correction models for rates

Strength: simple, robust, captures net new business

Weakness: mixes closures, new accounts and balance changes

Account level

Data: life and balance of every account

Methods: survival analysis, vintage decay, balance per surviving account

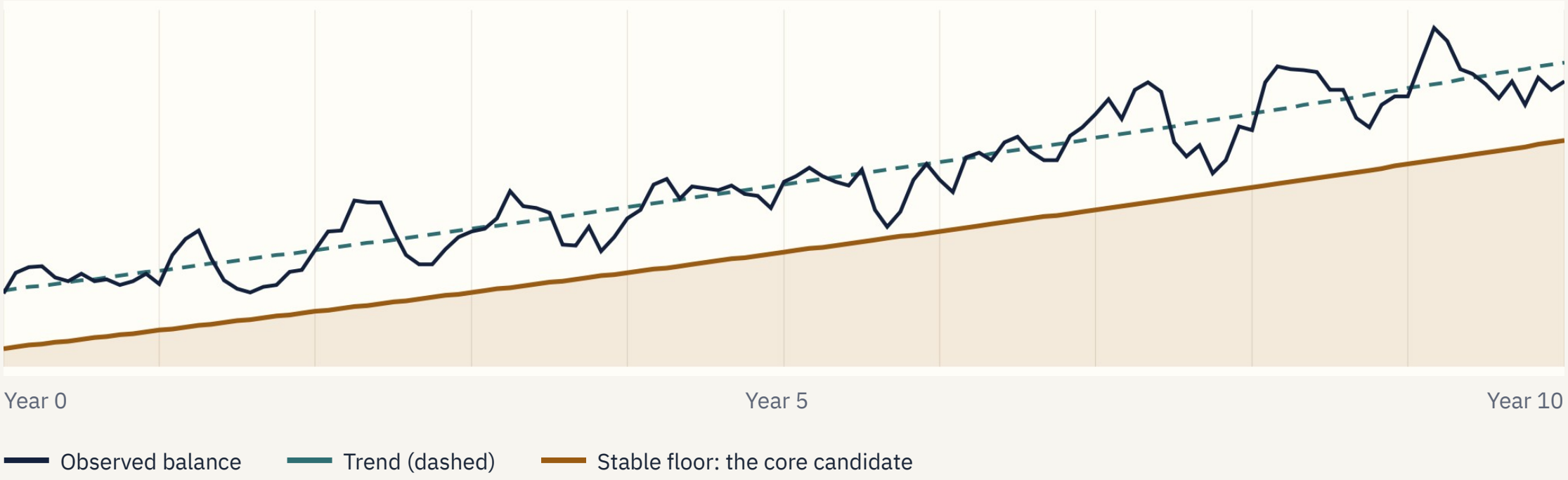
Strength: separates attrition from balance dynamics; natural run-off view

Weakness: data-hungry; misses money moving between accounts

> EVE needs a run-off view (no new business); the NII test needs a constant balance sheet. One model must serve both.

Separate trend, seasonality and noise

$$\ln B(t) = trend(t) + season(t) + \varepsilon(t)$$



PART 6 · VOLATILITY ENVELOPE

A worst-case floor for the balance

$$Core(h) = B(0) \cdot exp[\, min(\mu, 0) \cdot h - z(\alpha) \cdot \sigma \cdot \sqrt{h}\,]$$

With probability α, the balance stays above Core(h) after h months

μ

Monthly drift of ln B; no credit for growth

σ

Monthly volatility of changes in ln B, after removing seasonality

z(α)

Normal quantile: 2.33 at 99%

h

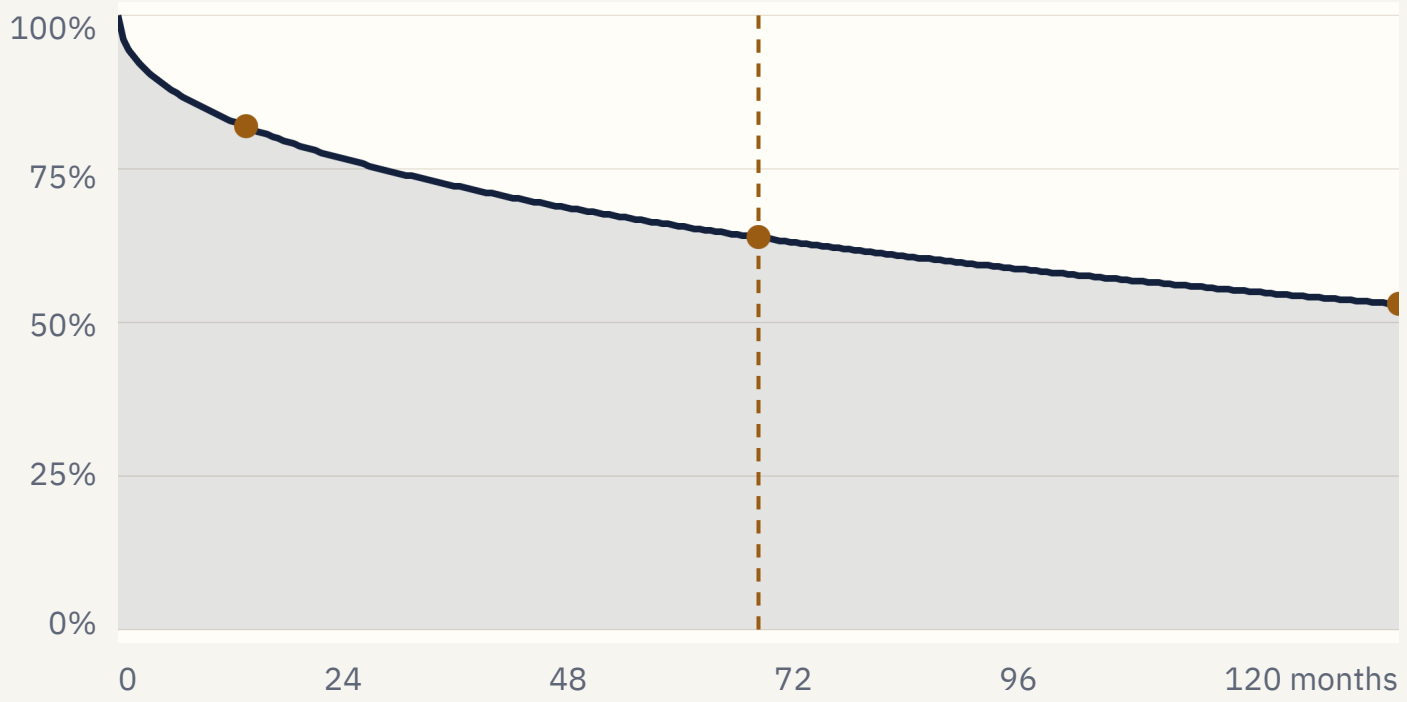
Horizon in months

> Amount leaving between h1 and h2 = Core(h1) − Core(h2): it is slotted in that time bucket.

Log-normal approximation; fat tails and autocorrelation call for a historical or bootstrap version · 19

PART 6 · ENVELOPE EXAMPLE

From envelope to repricing profile



Horizon	Stable share
1 month	94.4%
12 months	81.8%
24 months	75.2%
60 months	63.7%
120 months	52.9%

Residual parked at 10 years: average repricing **6.6 years**, above the 5-year cap

Horizon cut at 5 years: **3.7 years**

Illustrative: σ = 2.5% a month, μ = 0, α = 99% · volatile 5.6% slotted overnight · 20

PART 6 · RUN-OFF MODELS

Exponential decay of today's balances

$$B(t) = B(0) \cdot e^{(-\lambda t)}, \lambda = -\ln(1 - \text{annual attrition})$$

Run-off view: no new accounts, no new money

15%

annual attrition, so λ = 0.163

6.15 y

average life, 1 / λ

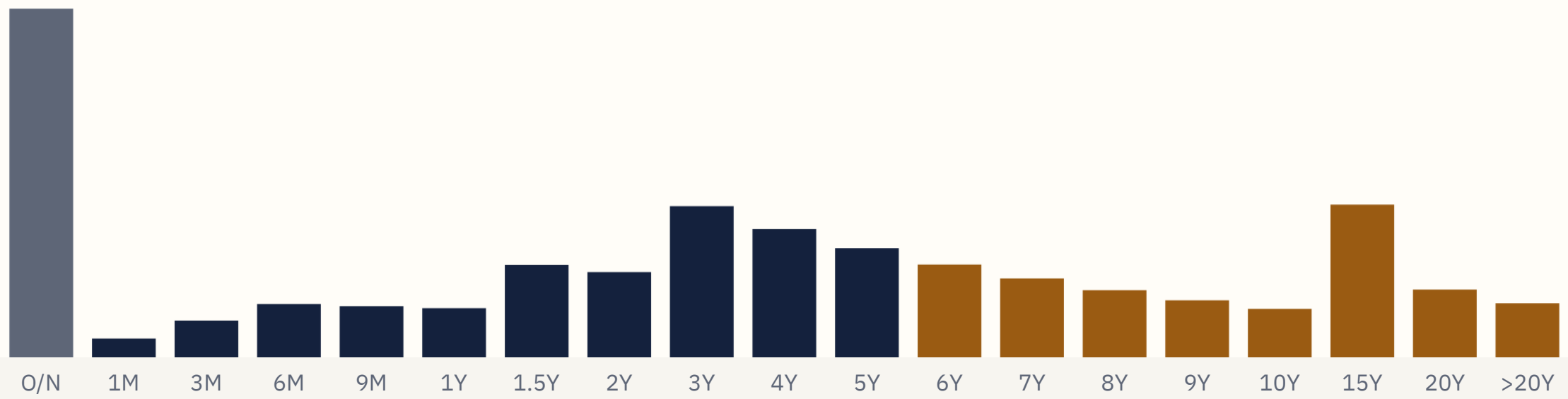
4.27 y

half-life, ln 2 / λ

> Estimate λ per segment and vintage: young accounts close faster, so the decay rate falls with account age.

PART 6 · DECAY EXAMPLE · €1,000M, 80% CORE, 15% ATTRITION

The profile across the 19 regulatory buckets



Gray: €200m non-core, overnight. Navy: core repricing within 5 years. Brown: beyond 5 years.

Portfolio average repricing: **4.73 years**, within the 5-year cap

Same model with 90% core: **5.32 years**, breaches the cap

Bar labels show each bucket's upper edge; buckets have unequal widths; balances beyond 20 years counted at 20 years · 22

PART 6 · SURVIVAL MODELS

Who leaves, and when

$$h(t \mid x) = h0(t) \cdot \exp(b1 \cdot spread + b2 \cdot age + b3 \cdot balance + \dots)$$

$$S(t) = \exp(-\int h(u)\, du),\ \text{expected run-off} = balance \times S(t)$$

Kaplan–Meier

Non-parametric survival curve by segment or vintage

Cox hazard

Covariates such as the rate spread shift the closure hazard

Competing risks

Closure, switch to term deposit, or slow run-down

> Accounts still open are censored, not dropped: dropping them overstates attrition.

PART 6 · JARROW AND VAN DEVENTER (1998)

Volume and rate as functions of the short rate

BALANCE

$$\ln D(t) - \ln D(t-1) = b0 + b1 \cdot r(t) + b2 \cdot [r(t) - r(t-1)] + b3 \cdot t + \varepsilon(t)$$

CLIENT RATE

$$i(t) - i(t-1) = a0 + a1 \cdot r(t) + a2 \cdot [r(t) - r(t-1)] + \eta(t)$$

FRANCHISE VALUE

$$V = E[\, \Sigma\, (\, r(t) - i(t) - c\,) \cdot D(t) \cdot DF(t)\,]$$

> Deposits are valued like an exotic swap: simulate rate paths, project D and i on each path, discount the spread earned.

Specification as commonly implemented; c = servicing cost · Journal of Banking and Finance 22 (1998) · 24

PART 6 · MIGRATION MODELS

Switching into term deposits

$$p(t) = 1 / (1 + exp[-(a + b \cdot s(t))]),\ s = \text{term rate} - \text{NMD rate}$$

p = share of NMD balances moving to term deposits per year

Spread s	Annual migration
0 pp	1.8%
1 pp	3.9%
2 pp	8.3%
3 pp	16.8%

Euro area, October 2023: households earned 0.35% on overnight deposits and 3.27% on new term deposits up to one year.

A spread of about **2.9 pp**: deep in the steep part of the curve.

Illustrative a = −4, b = 0.8 · Rates: ECB MFI interest rate statistics, release of 4 December 2023 · 25

PART 7 · RATE MODELS

Deposit beta: how much reaches the client

$$\text{cumulative beta}(t) = [\, i(t) - i(0) \,] / [\, r(t) - r(0) \,]$$

+450 bp

ECB deposit facility rate, Jul 2022 to Sep 2023

0.35%

household overnight deposit rate, Oct 2023

≈ 8%

implied cumulative beta on overnight deposits

32%

mean retail pass-through banks assume for +200 bp (EBA QIS)

Three different betas: long-run in the model, cumulative along a real cycle, and the scenario beta used in the outlier tests.

Sources: ECB MFI interest rate statistics (Dec 2023 release); EBA/REP/2025/04 · 26

PART 7 · ERROR-CORRECTION MODEL

Long-run target, short-run adjustment

$$\Delta i(t) = \gamma \cdot \Delta r(t) + \alpha \cdot [\, i(t-1) - c - \beta \cdot r(t-1) \,] + \varepsilon(t)$$

β

Long-run pass-through

example 0.30

γ

Immediate pass-through

example 0.05

α

Speed of adjustment, negative

example −0.10 a month

c

Level of the long-run spread

example −0.20%

Half-life of a gap

ln 0.5 / ln(1 + α) = 6.6 months in the example

Estimation

Test that i and r are cointegrated; estimate per segment on a sample with a full hiking cycle

PART 7 · THE MODEL UNDER THE OUTLIER-TEST SHOCKS

±200 bp through the error-correction model



Month	+200	−200
0	0.40%	0.40%
1	0.50%	0.30%
6	0.71%	0.10%
12	0.84%	0.00%
36	0.99%	0.00%

Cumulative beta after 12 months: **22%**; after 36: **29%**. Down shock: zero floor binds from month 10.

Parameters from slide 27 · market rate 2% before the shock · client rate floored at 0% · 28

PART 7 · BEYOND THE LINEAR MODEL

Asymmetry, floors and level-dependent betas

Asymmetry

Banks pass on rate rises more slowly than cuts. Estimate separate adjustment speeds for up and down moves.

Hannan and Berger (1991); Neumark and Sharpe (1992)

Zero floor

$$i(t) = max[\,0,\ model\ rate\,]$$

Near zero the beta collapses in the down direction and the margin is squeezed.

Level dependence

$$\beta(r) = \beta min + (\beta max - \beta min) / (1 + e^{(-k(r - r^*))})$$

Competition returns as rates rise, so the beta rises with the rate level.

> A beta estimated on 2012–2021 data, with rates near zero, says little about behavior at 3–4%.

PART 7 · THE ACADEMIC DEBATE

The deposit franchise: hedge or illusion?

HEDGE

Market power keeps deposit rates sticky. The franchise behaves like a long fixed-rate liability, so long fixed-rate assets are a hedge, not a bet.

Drechsler, Savov and Schnabl, Journal of Finance (2021)

ILLUSION

In 2022–23 asset losses were large, and the franchise is worth nothing to a bank whose uninsured depositors run.

Jiang, Matvos, Piskorski and Seru, Journal of Financial Economics (2024)

Regulatory reading: the franchise is real but conditional on stable volumes, a low beta and no run. The caps exist because those conditions can fail together.

PART 7 · PUTTING THE SUB-MODELS TOGETHER

One scenario, one consistent story



Same driver

Every sub-model reads the same rate path

Same direction

Higher beta and more migration go with less core, never more

Same limits

The 5-year cap holds in every scenario, not just the baseline

PART 8 · REPLICATING PORTFOLIOS

Invest the deposits in rolling tranches

$$y(t) = \Sigma\, w(k) \cdot MA[\, r(k), M(k)\,](t) \quad m(t) = y(t) - i(t)$$

Tranche k: tenor M(k), weight w(k), rolled monthly; its yield is the moving average of the M(k) rate over M(k)

Who uses it

Treasury and ALM, to set the internal transfer price of deposits

What it gives

A repricing profile: each tranche runs off evenly over its tenor

Average repricing

$\Sigma\, w(k) \cdot M(k) / 2$

MA = moving average · m = deposit margin · 32

Choosing the weights

$$\min_{w} : \mathrm{Var}[\, m(t) \,]$$

$$\text{subject to } \Sigma\, w(k) = 1,\ w(k) \geq 0,\ w(O/N) \geq \text{volatile share}$$

Other objectives

- Maximize mean margin per unit of margin volatility
- Minimize tracking error to the client rate
- Multistage stochastic programming with volume scenarios

Constraints that matter

- Volatile part held overnight
- Maximum tenor and maximum weight per tranche
- Average repricing within the 5-year cap

> The solution depends on the sample: in a falling-rate sample, long tenors look best because their yields lag.

PART 8 · REPLICATING PORTFOLIO EXAMPLE

Four tranches, one profile

Tranche	Weight	Tenor	Avg remaining
Overnight	30%	0	0
1 year	20%	1 y	0.5 y
5 years	30%	5 y	2.5 y
10 years	20%	10 y	5.0 y



1.85 y

average repricing: 0.2 × 0.5 + 0.3 × 2.5 + 0.2 × 5.0

10 y

longest repricing date, disclosed under CRR Art. 448

PART 8 · LIMITS OF REPLICATING PORTFOLIOS

What the optimizer does not tell you

1 **Wrong target**

It stabilizes the margin, not economic value. A stable margin can coexist with a large EVE exposure.

2 **Backward-looking**

Weights fitted on one rate regime. A long falling-rate sample favors long tenors.

3 **Volume blind**

Unless constrained, it assumes the balance is there to invest. Outflows break the ladder.

4 **Incentive to stretch**

Longer tenors earn carry today. EBA: hedging deposits is not yield optimization.

PART 8 · ASSEMBLING THE PROFILE

Seven steps from data to EVE and NII

1. **Segment** the book into homogeneous groups
2. **Volume model**: stable share and run-off, per scenario
3. **Rate model**: beta path; the rate-sensitive part of the stable balance is non-core
4. **Migration**: lower NMD volume in rising-rate scenarios
5. **Slot** the core along its run-off curve into the 19 buckets
6. **Check limits**: SA category caps, GL 5-year cap, internal limits
7. **Measure**: EVE on a run-off balance sheet, NII on a constant one

> Steps 2 to 7 are repeated for each of the six EVE scenarios and the two NII scenarios.

PART 8 · NII EXAMPLE · €1,000M NMD, 12 MONTHS, €M

How the investment choice shapes NII risk

Deposits invested in	Base	+200 bp	ΔNII	−200 bp	ΔNII
Overnight	16.0	33.0	+17.0	−1.1	−17.1
Half overnight, half 5-year ladder	16.0	24.1	+8.1	7.9	−8.1
5-year rolling ladder	16.0	15.2	−0.8	16.8	+0.8

Unhedged franchise

Wins when rates rise, loses the whole margin when they fall and the floor binds

Laddered franchise

NII barely moves; the rate risk has moved into EVE instead

Flat curve at 2%, instantaneous parallel shocks, constant balance sheet, client rate from the ECM on slide 28 · 37

PART 8 · EVE EXAMPLE · +200 BP PARALLEL, €M

What the NMD assumption is worth in EVE

Deposit assumption (€1,000m)	Gain on deposits	ΔEVE
All overnight, no behavioral model	0	−89
680 core spread over 6.4 years (average 3.2)	+37	−52
Same, core cut to 544 in the shock (0.8 scalar)	+22	−67

−89

asset side: €1,000m 5-year fixed bullet, 2% coupon

The behavioral model offsets about 40% of the asset loss.

The scenario scalar takes back more than 40% of that offset, because core shrinks exactly when rates rise.

Flat curve 2% to 4%, annual compounding, principal cash flows only · core profile from the worked example on slide 47 · 38

PART 9 · VALIDATION

Back-testing volume and rate models

Volume

- Count breaches of the predicted floor
- Compare predicted and realized run-off by vintage
- Test out of sample on the latest cycle

Rate

- Predicted vs realized client-rate path in 2022–24
- Error statistics: mean error and mean absolute error
- Re-estimate with and without the hiking cycle

⚠ The 2022–23 hiking cycle was the first true out-of-sample test for models calibrated on 2012–21. The EBA found many models were not recalibrated afterwards.

PART 9 · TESTING THE STABILITY FLOOR

Is the 99% floor really 99%?

$$LR = -2\,\ln[\,(1-p)^{(n-x)} \cdot p^{x}\,] + 2\,\ln[\,(1-x/n)^{(n-x)} \cdot (x/n)^{x}\,]$$

Chi-squared with 1 degree of freedom: reject the model if LR > 3.84 (5% level)

Breaches x	LR	Verdict
1	0.04	accept
2	0.45	accept
3	1.93	accept
4	4.10	reject
5	6.79	reject

n = 120 monthly observations, p = 1%: we expect 1.2 breaches.

Four or more breaches in ten years and the floor is too optimistic.

Kupiec (1995) proportion-of-failures test, as used for VaR back-testing · 40

PART 9 · SENSITIVITY ANALYSIS ON THE TOY BANK, €M

Which assumption moves the result

Assumption (low / base / high)	Low	Base	High	Metric
Core: 580 / 680 / 780	−57	−52	−46	ΔEVE, +200
Core maturity: 5.4 / 6.4 / 7.4 y	−57	−52	−47	ΔEVE, +200
Scalar on core: 0.8 / 1.0	−67	−52	n/a	ΔEVE, +200
Invested overnight: 100 / 50 / 0%	−17.1	−8.1	+0.8	ΔNII, −200

> Scenario dependency moves EVE more than a ±15% change in the core amount; for NII, the investment choice dominates.

Same assumptions as slides 37 and 38; maximum core maturity = twice the average · 41

PART 9 · MODEL GOVERNANCE

Governance for judgment-heavy models

Roles

Treasury builds, Risk validates independently, ALCO approves

Documentation

Data, segments, parameters, overrides and their rationale

Recalibration

At least yearly, and after any regime change in rates

Expert judgment

Written down, conservative, and reverse-stressed

Limits

GL cap plus internal limits on core share and maturity

Disclosure

Average and longest NMD repricing in Pillar 3

Basel Principle 5; EBA/GL/2022/14 governance section; CRR Art. 448 · 42

PART 10 · RTS 2024/857 · ART. 1 AND 8(1)

Four categories, defined by client and use

Retail transactional

Salaries, income or expenses regularly credited and debited, or no interest even when rates are high

Retail non-transactional

Not a transactional account, or interest-bearing

Wholesale non-financial

Corporates and other non-financial clients that are not retail

Wholesale financial

Always treated as non-core: no behavioral maturity

> **Core component:** the part of a stable NMD that is unlikely to reprice, even under significant changes in the rate environment.
>
> Retail follows CRR Art. 411(2): natural persons and SMEs within the retail threshold.

Source: Commission Delegated Regulation (EU) 2024/857, Art. 1(8)–(16), Art. 8(1) and 8(7) · 43

PART 10 · RTS 2024/857 · ART. 8

The standardized NMD algorithm in six steps

1 · Classify

Four categories by client and use

Art. 8(1)

2 · Stable share

From volume changes vs rate moves over at least 10 years

Art. 8(2)(a)

3 · Core share

Non-core of stable = stable × pass-through rate

Art. 8(2)–(3)

4 · Scalars

Core × 0.8 if short rates rise, × 1.2 if they fall

Art. 8(4)–(5)

5 · Caps

Core share at most 90% / 70% / 50%

Art. 8(6)

6 · Slot

Non-core overnight; core from internal data, average maturity at most 5 / 4.5 / 4 years

Art. 8(8)–(9)

De minimis (Art. 8(10)): NMDs under 2% of banking-book liabilities are all non-core · RTS 2024/857 · 44

PART 10 · SCENARIO DEPENDENCY

Core deposits shrink when short rates rise

SHORT RATES UP

× 0.8

Parallel up, flattener, short rates up. Clients chase yield: less core, more overnight.

SHORT RATES DOWN

× 1.2

Parallel down, steepener, short rates down. Fewer alternatives: more core, longer profile.

> The category caps (90% / 70% / 50%) still bind after the scalar is applied (Art. 8(6)).

Source: RTS 2024/857 Art. 4 and Art. 8(4)–(6) · 45

Small banks get prescribed core shares

Category	Baseline	Rates down	Rates up	Max maturity
Retail transactional	69.23%	90.00%	48.46%	5 years
Retail non-transactional	53.85%	70.00%	37.69%	4.5 years
Wholesale non-financial	38.46%	50.00%	26.92%	4 years

The arithmetic

Baseline = cap ÷ 1.3; down = baseline × 1.3 = the cap; up = baseline × 0.7

The slotting

Core spread evenly over time up to a maximum, not an average, maturity (Annex, point 5)

Core component as a share of each category · Source: RTS 2024/857 Art. 25(2)–(4) and Annex · 46

A €1,000m current-account book

Step (€m)	Baseline	Short rates up	Short rates down
Stable part (85%, estimated)	850	850	850
Non-core of stable (pass-through 20%)	170	170	170
Core before scalar	680	680	680
Scalar	× 1.0	× 0.8	× 1.2
Core after scalar	680	544	816
Cap check: 90% = 900	within	within	within
Non-core, slotted overnight	320	456	184
Core average maturity (assumed)	3.2 y	3.2 y	3.2 y
Portfolio average repricing (GL cap 5 y)	2.18 y	1.74 y	2.61 y

Illustrative figures · portfolio average = core × 3.2 ÷ 1,000, non-core at ~0 years · 47

PART 11 · RTS 2024/856 · CRD ART. 98(5)

Two supervisory outlier tests

ECONOMIC VALUE OF EQUITY

−15% of Tier 1

- Six shock scenarios
- Run-off balance sheet
- Margins per internal approach
- Risk-free discounting

NET INTEREST INCOME

−5% of Tier 1

- Parallel up and parallel down
- One year, constant balance sheet
- Commercial margins included
- New business at recent margins

Both: behavior adjusted to each scenario · EUR shocks 200 / 250 / 100 bp (parallel / short / long) · post-shock floor −150 bp, rising 3 bp a year to 0% at 50 years

Source: RTS 2024/856 Art. 1–5 and Annex; EBA Opinion EBA/Op/2023/03 · 48

PART 11 · WHERE NMD ASSUMPTIONS BITE

EVE fears rates up; NII fears rates down

Parallel up · EVE

Long-modeled deposits gain value and offset losses on fixed-rate assets. If deposits are really shorter, the true loss is larger.

Parallel down · NII

Client rates near zero cannot fall much further while asset yields do: the deposit margin is squeezed.

0.66% vs 7.24%

Share of banks flagged as outliers on EVE vs NII (EBA, Jan 2026)

0 → 40%

EVE outliers end-2023 with vs without interest rate derivatives (EBA, Feb 2025)

Sources: EBA/REP/2025/04; EBA second heatmap report (Jan 2026) via msg for banking · 49

PART 11 · NMD MARGINS IN THE NII OUTLIER TEST

Same margin logic as the internal model

Internal model has scenario margins

Apply them in the SOT on NII, conservatively

No margin model

Constant spread over the risk-free rate in every scenario

Practice, Dec 2023

38% used constant NMD margins, 62% did not

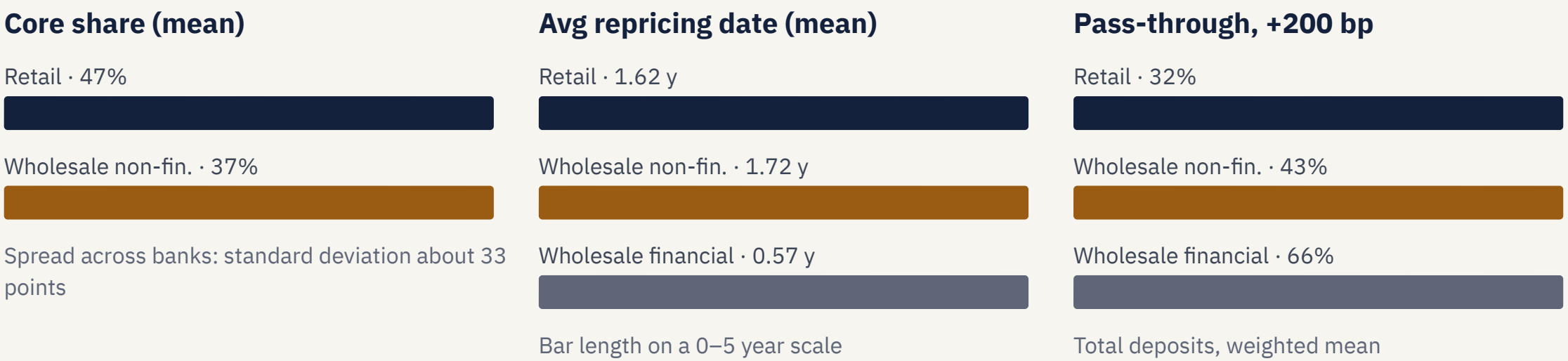
Circumstances the EBA asks modelers to reflect

Situation	Prudent modeling consequence
Large negative spread to risk-free	Model margin compression: it may erode over time
Risk-free rate at or below the down shock	Assume the NMD margin is extinguished in the down scenario
Recent rate rises, lagged pass-through	Model a catch-up in client rates, i.e. spread compression

Source: EBA/REP/2025/04 section 1.5; RTS 2024/856 Art. 4(4) · 50

PART 12 · WHAT EU BANKS ACTUALLY ASSUME

Far inside the caps, and widely dispersed



Source: EBA/REP/2025/04, Tables 5–7, QIS data as of December 2023 · 51

PART 12 · EBA LIST OF NMD RISK FACTORS

What drives deposit behavior

Customer

- Transactional or not
- Covered by deposit guarantee or not
- Balance size, income, age, loyalty
- Digital interaction, advisory
- Wholesale: sector, size, excess cash

Institution

- Size and market position
- Deposit concentration
- Liquidity and credit rating
- Business and funding model
- Services and reputation

Market

- Competition
- Market liquidity and concentration
- Alternative investments and rate gaps
- Costs of deposits
- Digital and tech innovation

Economy

- Policy and regulation
- Seasonality and the cycle
- Inflation, GDP, unemployment
- Interest rates, credit spreads
- Health of the banking system

PART 12 · EBA SUPERVISORY TOOLKIT

How supervisors will challenge your model

Segmentation and peers

Compare repricing maturity, core share, pass-through and migration across banks with similar risk factors

Stress and reverse stress

Which NMD assumptions would breach the SOT or an internal limit?

History and judgment

Balance historical data with forward-looking analysis and expert judgment

Expected directions

A starting list of how each risk factor should move core share and maturity

Source: EBA/REP/2025/04, section 1.3.2 · also feeds the EBA's update of the SREP framework · 53

Sanity checks any NMD model should pass

When this increases	Non-core share	Average maturity
Current and projected cost of NMDs	Higher	Shorter
Share of non-remunerated NMDs (high rates)	Lower	Longer
Share of retail transactional NMDs	Lower	Longer
Minimum historical balance per account	Lower	Longer
Concentration by depositor	Higher	Shorter
Bank's concentration in its market	Lower	Longer
Deposit size, client wealth or income	Higher	Shorter

PART 12 · CASE STUDY

2023: two very different stress tests

UNITED STATES · SILICON VALLEY BANK

$42bn

withdrawn on 9 March 2023, with more than $100bn more expected the next day

Concentrated, uninsured, digitally connected depositors: a volume shock in hours

EURO AREA · SYSTEM-WIDE

0.35% vs 3.27%

household overnight vs new term deposits up to one year, October 2023

No run, but migration into term deposits and slowly rising betas: a rate and mix shock

> Insured share, concentration and channel decide whether stress shows up as outflows or as repricing.

Sources: M. Barr, testimony to the US Senate Banking Committee, 28 Mar 2023; ECB MFI interest rate statistics · 55

PART 13 · HEDGING NMDS

Model the deposits, then hedge them, never the reverse

Hedging NMDs should target prudent IRRBB management, not yield optimization.

Derivatives are for hedging, not speculation.

Do not assume a natural hedge with the repricing of loans.

NMD repricing must be modeled on the deposits' own features.

Interest rate swaps are the main tool, after natural hedges

Deposits are mostly hedged with portfolio (macro) fair value hedges

Some banks build swap portfolios as a structural hedge of core deposits

Source: EBA/REP/2025/04, section 1.6 · 56

An economic hedge that hurts the P&L

Core deposits

Measured at amortized cost: their economic gain when rates rise is invisible in the accounts

Receive-fixed swaps

Derivatives at fair value through profit or loss: every rate move hits the P&L

WITHOUT HEDGE ACCOUNTING

−€46m

P&L on a €1bn 5-year receive-fixed swap at 2% when rates rise by 100 bp, with no offset recognized on the deposits

> Hedge accounting exists to remove this mismatch. For core deposits, full IFRS does not allow it; the EU version does.

Illustrative: annual fixed leg repriced at 3%, floating leg at par · 57

PART 13 · IAS 39 MACRO HEDGE (2004 AMENDMENT)

Portfolio fair value hedge of interest rate risk

1 · Portfolio

Identify a portfolio of assets, liabilities or both exposed to interest rate risk

2 · Buckets

Allocate it to repricing periods on expected, not contractual, dates

3 · Amount

Designate an amount of assets or liabilities in each period as the hedged item

4 · Derivatives

Designate interest rate swaps as hedging instruments

5 · Remeasure

Fair value change of the hedged amount, for the hedged rate, shown on a separate balance-sheet line

6 · Test

Effectiveness within 80–125%; the rest is ineffectiveness in P&L

Why core deposits cannot be the hedged item

> *"The fair value of a financial liability with a demand feature (eg a demand deposit) is not less than the amount payable on demand, discounted from the first date that the amount could be required to be paid."*
>
> IAS 39.49, now IFRS 13.47

No fair value hedge

Fair value does not move with rates, so there is nothing to hedge beyond the demand date

No cash flow hedge

Interest on core deposits does not vary with market rates

Revisions cost

Revising expected repricing dates creates ineffectiveness

PART 13 · COMMISSION REGULATION (EC) NO 2086/2004

The EU carve-out: two relaxations

1

Core deposits as hedged items

Core demand deposits, slotted on behavioral repricing dates, can be designated in a portfolio fair value hedge, including their interest rate component when they pay zero or below-market rates

2

Bottom layer, no revision noise

Revised expectations in a time bucket create no ineffectiveness as long as the bucket still holds at least the hedged amount (under-hedging)

Deletions in IAS 39 paras 81A, AG99C, AG99D, AG107A, AG114, AG118, AG119, AG121, AG122, AG124, AG126, AG127, AG129, AG130

From behavioral model to hedge designation

Core profile per bucket from the IRRBB model → Bottom layer per bucket, with a buffer → Receive-fixed swaps designated → Monthly re-estimate and test

Bucket still covers the layer

No ineffectiveness from the revision; the hedge continues

Bucket falls below the layer

Over-hedge: ineffectiveness on the excess, and rebalance or de-designate

> Auditors expect breaches to be tracked by generation of hedges, using the same model as IRRBB.

PART 13 · CARVE-OUT EXAMPLE · 3–4 YEAR BUCKET, RATES +100 BP, €M

A bucket that holds, and one that breaches

Modeled core deposits: **200**

Designated bottom layer: **150**

Receive-fixed swaps: **150**

Case	Deposits	Hedged item	Swaps	P&L
A · no revision	200	+4.7	−4.7	0.0
B · revised to 170	170	+4.7	−4.7	0.0
C · revised to 120	120	+3.8	−4.7	−0.9
No hedge accounting	any	0.0	−4.7	−4.7

Case B is the carve-out at work: a 15% downward revision, no P&L. Case C: only the 30 of over-hedge is ineffective.

Fair value changes on a 3.5-year zero-coupon proxy: 3.13 per 100 for +100 bp from 2% · 62

PART 13 · THE CARVE-OUT UNDER IFRS 9

IFRS 9 did not end the carve-out

IFRS 9 para 6.1.3

For a fair value hedge of the interest rate exposure of a portfolio, an entity may keep applying the IAS 39 requirements

IFRS 9 para 7.2.21

An accounting policy choice to keep IAS 39 hedge accounting altogether

In the EU

"IAS 39 as endorsed" includes the carve-out, so it remains available to banks

In practice

EBA (2025): deposits are mostly hedged through portfolio fair value hedges

> The IASB kept these options open until its macro-hedging project is finished.

PART 13 · WHAT COMES NEXT

IASB Risk Mitigation Accounting

3 Dec 2025

exposure draft, amendments to IFRS 9 and IFRS 7

31 Jul 2026

end of the 240-day comment period

TBD

effective date, after redeliberations

The model

- Net repricing risk by time band, on expected dates
- Stable core demand deposits can count as fixed-rate liabilities
- A risk mitigation objective, mirrored by benchmark derivatives
- Adjustment = lower of derivative gains or losses and benchmark change

For EU banks

IAS 39 macro hedges would be discontinued on transition. Whether the carve-out survives depends on EU endorsement.

Sources: IASB; BDO, Forvis Mazars and EY summaries of the ED (Dec 2025) · status to be checked before the lecture · 64

PART 14 · OPEN DEBATES

Where the rules are still contested

Are the caps too tight?

An Italian industry response to the SA consultation cited estimated NMD lives above nine years, against 5 / 4.5 / 4-year limits

Keep the 5-year cap?

Adopted for a period of sharp rate rises; in 2026 the EBA found it harmonizing and recommended keeping it

New shock sizes

Basel 2024 recalibration: local shock factors, 99.9th percentile, 25 bp rounding. EU implementation still to be decided.

Carve-out or new standard?

IASB Risk Mitigation Accounting could replace IAS 39 macro hedges; the EU must decide whether to keep its carve-out

Sources: EBA consultation responses on the SA RTS; BCBS d578 (Jul 2024); EBA heatmap reports 2025–2026; IASB ED (Dec 2025) · 65

PART 14 · KEY TAKEAWAYS

Seven things to remember

1. Behavioral modeling is allowed if sound, documented, tested and scenario-dependent.
2. Internal models: NMD average repricing date capped at 5 years (GL para 111).
3. Volume, rate and migration models must tell one story per scenario.
4. SA: core caps 90 / 70 / 50%, maturity caps 5 / 4.5 / 4 years, scalars 0.8 / 1.2.
5. Outlier tests: EVE −15% and NII −5% of Tier 1; NMDs drive both.
6. Supervisors benchmark peers, reverse-stress assumptions and check directions.
7. The EU IAS 39 carve-out lets banks hedge-account core deposits.

PART 14 · DISCUSSION AND EXERCISE

Your turn

Q1 A bank with mostly non-remunerated salary accounts says the 5-year cap is too strict. What evidence would you ask it to provide?

Q2 A pass-through model calibrated on 2012–2021 data gives a beta of 10%. How would you challenge it?

Q3 How can NMD assumptions make a bank look safe on EVE and risky on NII at the same time?

EXERCISE · 5 MINUTES

Re-run the €1,000m example with a stable share of 70% and a pass-through rate of 45%.

Core in each scenario? Portfolio average repricing date with a 3.2-year core maturity? Which assumption moved the result most?

PART 14 · EXERCISE ON THE MODELS

Calibrate, slot, stress, account

1 **Envelope.** Monthly volatility 3%, no drift, 99%. What share of the balance is stable at 12 and at 60 months?

2 **Decay.** Annual attrition 20%. Average life and half-life? With 85% core, truncated at 20 years, is the 5-year cap met?

3 **Error correction.** $\beta = 0.5$, $\gamma = 0.1$, $\alpha = -0.2$, $c = 0$. Start at $r = 2\%$, $i = 1\%$; rates jump +200 bp. Client rate after 1, 2 and 3 months?

4 **Carve-out.** A bucket holds 90 against a layer of 100. What is ineffective, and what should Treasury do?

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